

GREEN TECH IS DOOMED

Read on to find out how ignorance, fatalistic tendencies and irresponsibility will inevitably destroy the human race

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We know that traditional green tech will not save the world any time soon. We can fool ourselves into thinking otherwise, but the truth is that it's just not enough. But why is green tech doomed? Wouldn't a massive global effort to restructure our entire power infrastructure have an impact? What if everyone starts using solar chargers and electric cars and public transport and the like? How will it all add up?

The math

The answer to those above questions is yes, there will be an impact but green tech's contribution to global power production right now is so irrelevant that it just can't be enough to have a truly major impact. Just as an example, have a look at the following table indicating the contribution of solar and wind power in the U.S. in 2010 and as you can see, the shortfall is simply staggering. Without going into the nitty-gritty of things, even if the U.S. were to increase the number of generators by 200 times, the actual contribution would only be close to 1% of the total electricity production in the U.S. This is without even considering the amount of investment in terms of land, time and capital and completely ignoring

the predicted annual increase in the demand for electricity. The only relevant green power source that we actually have right now is hydroelectric power. That is one of the best examples of green energy done right; harnessing the potential energy of entire rivers to produce power for the masses. Four of the top five power generators are hydroelectric power plants and while this seems like a very good thing for green tech in general, these are not structures that one can build anywhere. They are extremely niche and are entirely dependent on luck and location. If you look at hydroelectricity as a whole, all the hydroelectric power plants in the world account for barely 16% of the world's electricity production. This is not a value that can be dramatically improved upon just because we feel like it. There can be incremental updates, improvements in efficiency, etc., but all that will do is maintain that 16% hold on world power generation. Solar and wind power are in a more woeful state. Simply put, one square kilometre of solar panels are capable of generating only 60MW of power, say you double the efficiency using some magical process that hasn't yet been discovered (or

has it?), you'll still get only around 100-150MW per square kilometre. In contrast, one square kilometre of nuclear power station can produce almost 2000MW of power. Thermal power plants are not very far behind either. Of course, coal-fed plants generate tremendous amounts of CO₂, the largest of which (situated in Taichung, China) generates 40 million tonnes of CO₂ annually. Of course these stations are harmful, but they also produce ridiculous amounts of power. Green tech just cannot hope to match up to that under any circumstances. For countries where space comes at a premium, even considering large-scale green energy solutions is simply out of the question. When a tiny little generator in a shop generate can generate more power than that same shop festooned with solar panels, how can any traditional green solution even hope to compete? People would rather pay triple the price for one litre of petrol than invest lakhs into a long-term solution that is entirely dependent on the whims of the weather. This brings us up very neatly to the next major issue with green energy. Traditional green energy such as wind, solar or even tidal power is entirely dependent on their location and the suitability of a region to support that kind of power. A rainy day might cause power output to drop drastically and a still night will bring entire wind farms to a halt. Reliability is not guaranteed and thus,

Power Source	Number of generators	% annual contribution
Wind	781	2.9
Solar	326	0.04

Source: Wikipedia